

Emergence of Spatial Locality from Information-Theoretic Constraints: The Critical Role of Finite Bandwidth

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Abstract

We investigate whether spatial locality can emerge from purely information-theoretic constraints on Hilbert space dynamics. Initial numerical experiments with three constraints (no-signaling, no-forgetting, no-refolding) showed negative results: coarse-grained factorizations consistently outperformed spatial locality by pooling more degrees of freedom. However, upon including a fourth constraint—finite bandwidth—the situation reverses dramatically. We demonstrate that the four constraints together create a tension that only spatial locality can resolve. Numerical simulations on systems up to $N = 24$ qubits ($2^{24} \approx 1.7 \times 10^7$ dimensional Hilbert space) show that spatial factorization wins decisively, with the selection margin *increasing* with system size. This provides evidence that classical spatial structure may be an inevitable consequence of information-theoretic constraints rather than a contingent feature of our universe.

1 Introduction

The tensor product structure of Hilbert space is typically assumed rather than derived. Why does the universe admit a preferred factorization corresponding to spatially local degrees of freedom? The Hilbert Substrate Framework attempts to answer this question by identifying minimal constraints that any physical theory must satisfy, and testing whether these constraints select for spatial structure.

This note reports results from a systematic numerical investigation. We find that three constraints are insufficient, but four constraints—including finite bandwidth—do select for spatial locality.

2 The Four Constraints

The framework imposes four regulative constraints on Hilbert space dynamics:

1. **No-signaling:** Influence propagates at finite speed. Operations on one subsystem cannot instantaneously affect distant subsystems.
2. **No-forgetting:** Information is conserved under unitary evolution. Correlations may delocalize but cannot be destroyed.
3. **No-refolding:** Once record-bearing structure exists, the factorization cannot be arbitrarily coarsened without destroying those records.

4. **Finite bandwidth:** Each subsystem has limited information processing capacity. A block of k qubits with bandwidth capacity B can only effectively utilize a fraction $\min(k, B)/k$ of its potential information content.

The key insight is that these constraints create *tension*:

- No-forgetting favors coarse graining (larger blocks retain more signal)
- Finite bandwidth favors fine graining (smaller blocks stay within capacity)
- No-refolding requires sufficient structure (cannot trivialize)
- No-signaling requires causal propagation paths

3 Methodology

We evaluate different tensor factorizations (“blockings”) of an N -qubit system by scoring how well each satisfies the four constraints. For a blocking with block size b :

- **Recoverability:** Can a local perturbation be detected at remote blocks after time evolution? Measured via trace distance of reduced density matrices.
- **Bandwidth efficiency:** $\eta = \min(b, B)/b$ where B is the bandwidth capacity.
- **Effective recoverability:** Raw recoverability $\times$ efficiency. This is the key metric—coarse blocks may have high raw signal but low effective signal due to bandwidth limitations.
- **Structure penalties:** Factorizations with too few blocks or degenerate block graphs are penalized.

The composite score rewards effective recoverability and penalizes constraint violations:

$$\text{Score} = 2.0 \times f_{\text{recover}}^{\text{eff}} + 0.8 \tanh\left(5 \times \bar{r}^{\text{eff}}\right) + 0.5 \times f_{\text{reached}} - \text{penalties} \quad (1)$$

4 Results

4.1 Three Constraints: Negative Result

With only three constraints (no-signaling, no-forgetting, no-refolding), coarse-grained factorizations consistently win. At $N = 12$:

Block Size	# Blocks	Raw Recoverability	Score
1 (spatial)	12	0.082	+0.764
2	6	0.945	+2.535
3	4	1.000	+2.729 ← winner
4	3	0.000	−1.500

The intermediate blocking ($b = 3$) wins because it pools enough qubits to maintain strong correlations while avoiding structure penalties. *Spatial locality is not selected.*

4.2 Four Constraints: Positive Result

Adding the finite bandwidth constraint ($B = 1.5$ qubits) dramatically changes the outcome. At $N = 24$ (2^{24} dimensional Hilbert space):

Block	# Blocks	Efficiency	Raw Recov.	Eff. Recov.	Score
1	24	1.00	0.80	0.80	+1.995 ← winner
2	12	0.75	0.80	0.60	+1.268
3	8	0.50	0.60	0.00	−0.647
4	6	0.38	0.80	0.00	−1.216
6	4	0.25	0.00	0.00	−2.478

Spatial locality wins decisively. The mechanism is clear:

- Coarse blocks ($b = 3, 4$) have excellent raw recoverability (60–80%)
- But their low bandwidth efficiency ($< 50\%$) kills effective recoverability
- Only spatial blocks ($b = 1$) have 100% efficiency *and* good recoverability

4.3 Scaling with System Size

The result is robust across system sizes, with the selection margin *increasing*:

N	Hilbert Dimension	Spatial Score	Next Best
12	4,096	+0.571	+0.140
20	1,048,576	+0.344	+0.105
24	16,777,216	+1.995	+1.268

At $N = 24$, spatial locality wins by a margin of +0.73, larger than at smaller N .

4.4 Phase Transition in Bandwidth

We observe a phase transition as bandwidth capacity varies. At $N = 20$:

Bandwidth B	Winner	Interpretation
0.5	$b = 1$ (spatial)	Severely constrained
1.0	$b = 1$ (spatial)	Constrained
1.5	$b = 1$ (spatial)	Critical regime
2.0	$b = 2$ (intermediate)	Relaxed
3.0	$b = 2$ (intermediate)	Unconstrained

The critical bandwidth is $B_c \approx 1.5$ –2.0 qubits. Below this threshold, spatial locality is the unique solution to the constraint tension. Above it, some coarse-graining becomes viable.

5 Discussion

5.1 The Essential Role of Finite Bandwidth

The finite bandwidth constraint is not merely an additional penalty—it fundamentally changes the optimization landscape. Without it, the no-forgetting constraint creates a monotonic preference for coarse graining: bigger blocks always see more signal. With finite bandwidth, this preference is bounded: blocks larger than the capacity suffer diminishing returns.

The physical interpretation is natural: any information-processing system has finite capacity. A subsystem cannot track arbitrarily many correlations simultaneously. This limitation forces the universe to adopt a factorization where each subsystem stays within its processing budget—and spatial locality is the optimal solution.

5.2 Why Spatial Locality Uniquely Satisfies All Four Constraints

1. **No-signaling:** Spatial locality defines a natural light-cone structure. Information propagates through neighbor-to-neighbor interactions.
2. **No-forgetting:** Although individual qubits have weak signal, the spatial structure preserves correlations through the interaction graph. Information spreads but remains recoverable.
3. **No-refolding:** With N blocks, the factorization has maximal structure. No coarsening is possible without losing spatial resolution.
4. **Finite bandwidth:** Single-qubit blocks have 100% efficiency. They never exceed capacity, so all detected signal contributes to the score.

Coarse-grained factorizations fail on point 4: they may satisfy the first three constraints but cannot stay within bandwidth.

5.3 Implications for Physics

If the four constraints are physically fundamental, then spatial locality is not a contingent feature of our universe but an *inevitable* consequence of information-theoretic limitations. The three-dimensional structure of space may similarly be constrained (see Paper II on accessibility collapse and the “window of reality”).

This suggests a research program: derive the dimensionality of space, the form of physical laws, and the emergence of particles from the interplay of information-theoretic constraints on Hilbert space.

6 Conclusion

We have demonstrated that four information-theoretic constraints—no-signaling, no-forgetting, no-refolding, and finite bandwidth—together select for spatial locality in Hilbert space factorization. The result is robust from $N = 12$ to $N = 24$ qubits, with the selection margin increasing at larger system sizes.

The critical finding is that *three constraints are insufficient*. Without finite bandwidth, coarse-grained factorizations win by pooling degrees of freedom. The fourth constraint creates the necessary tension: only spatial locality can satisfy all four simultaneously.

This provides numerical evidence that classical spatial structure may emerge inevitably from information-theoretic principles, rather than being assumed as a primitive feature of physical theory.

Acknowledgments

Numerical simulations used Trotter decomposition for time evolution, enabling exact treatment of systems up to $N = 24$ qubits (2^{24} -dimensional Hilbert space).